# PH3103 (Quantum Mechanics)

## Tutorial - 8

① (a) ~~Let~~ ~~$|\psi\rangle$ be the stationary state ket~~

(i) To ~~$\hat{x}|\psi\rangle$~~ $\hat{x}|\psi\rangle$ ~~$\rightarrow$~~ ~~$\pi|\psi\rangle$~~

$$\langle x|\pi|\psi\rangle = \langle x|\pi|\left(\int_{-\infty}^{\infty} dx' \, |x'\rangle\langle x'|\right)\psi\rangle$$

$$= \int_{-\infty}^{\infty} dx' \, \langle x|\pi|x'\rangle\langle x'|\psi\rangle$$

$$= \int_{-\infty}^{\infty} dx' \, \langle x|-x'\rangle\langle x'|\psi\rangle$$

$$= \int_{-\infty}^{\infty} dx' \, \delta(x+x')\langle x'|\psi\rangle = \langle -x|\psi\rangle$$

$$= \psi(-x)$$

$$\boxed{\langle x|\pi|\psi\rangle = \psi(-x)} \quad \text{(Proved)} \quad \text{— ①}$$

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}$$

$$\therefore \langle x|\pi|p\rangle = \langle -x|p\rangle$$

$$= \frac{1}{\sqrt{2\pi\hbar}} e^{ip(-x)/\hbar}$$

$$= \frac{1}{\sqrt{2\pi\hbar}} e^{i(-p)x/\hbar} = \langle x|-p\rangle$$

$$\therefore \langle x|\pi|p\rangle = \langle x|-p\rangle \Rightarrow \boxed{\hat{\pi}|p\rangle = |-p\rangle} \quad \text{(Proved)}$$

— ⓜ

(ii) $\pi|\psi\rangle = \lambda|\psi\rangle \; ; \; \lambda = \pm 1$.

$\lambda = 1$ : $\pi|\psi\rangle = |\psi\rangle \Rightarrow \langle x|\pi|\psi\rangle = \langle x|\psi\rangle = \psi(x)$

$\Rightarrow \boxed{\psi(-x) = \psi(x)}$ [from ①]

$\Rightarrow \boxed{\Pi^{+} \hat{p} \Pi = -\hat{p}}$ (Proved)

(b) (ii) $\Pi^{+} x \Pi = -x$

$\Rightarrow \Pi(\Pi^{+} x \Pi) = -\Pi x$

$\Rightarrow x\Pi = -\Pi x \quad [\Pi \Pi^{+} = \mathbb{I}]$

$\Rightarrow x\Pi + \Pi x = 0 \Rightarrow \boxed{[\Pi, x]^{+} = 0}$ (Proved)

$\Pi^{+} p \Pi = -p \Rightarrow p\Pi = -\Pi p$

$\Rightarrow \boxed{[\Pi, p]^{+} = 0}$ (Proved)

(iii) $\Pi^{+} L \Pi = \Pi^{+} (x \times p) \Pi$

$= (\Pi^{+} x \Pi) \times (\Pi^{+} p \Pi)$

$= (-x) \times (-p) = x \times p = L.$

$\therefore \Pi^{+} L \Pi = L$

$\Rightarrow L\Pi = \Pi L \Rightarrow \Pi L - L\Pi = 0$

$\Rightarrow \boxed{[\Pi, L] = 0}$ (Proved)

(2) $\frac{dp}{dt} = \frac{1}{i\hbar}[p, H]$ — Heisenberg's EOM.

$[p, H] = [p, \frac{p^2}{2m} + V(x)] = [p, V(x)]$

$\cancel{= -i\hbar \frac{\partial V}{\partial x} +}$

$\cancel{[p, V(x)]}$

$[p, V(x)]\psi = -i\hbar \frac{\partial}{\partial x}(V\psi) + V i\hbar \frac{\partial \psi}{\partial x}$

$= -i\hbar \frac{\partial \psi}{\partial x} \cdot V - i\hbar \frac{\partial V}{\partial x}\psi + V i\hbar \frac{\partial \psi}{\partial x}$

$= -i\hbar \frac{\partial V}{\partial x} \cdot \psi.$

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$$\therefore \quad [p, V(x)] = -i\hbar \frac{\partial V}{\partial x} = -i\hbar \cdot V_0 \cdot \frac{2\pi}{a} \cos\left(\frac{2\pi x}{a}\right)$$

$$[p, V(x)] = \frac{-i\hbar V_0 2\pi}{a} \cos\left(\frac{2\pi x}{a}\right).$$

$$\therefore \quad [p, H] = \frac{-2\pi i\hbar V_0}{a} \cos\left(\frac{2\pi x}{a}\right)$$

$$\therefore \quad \frac{dp}{dt} = \frac{1}{i\hbar}[p, H] = \frac{-2\pi V_0}{a} \cos\left(\frac{2\pi x}{a}\right) \neq 0.$$

Hence, p is not conserved.